

Assignment #2 – Student Project

Proposal Deadline: Thursday, April 30 – 23:59 (via Teams)

Final Submission: Monday, June 1 – 23:59 (via GitHub Classroom)

GitHub Classroom Link: <https://classroom.github.com/a/Jl7Pxp0W>

Deadlines are hard deadlines! Make sure to push your progress regularly to Github.
(No “*My dog ate my computer*”)

Assignment Overview

For Assignment #2, you will **design and develop your own Python-based application** using concepts and technologies learned in class. You are free to choose your topic (e.g., game, social app, utility, business tool), but you must submit a project proposal for approval before starting development. You may use the

Proposal Requirements (Mandatory submission)

Submit a **2-3 page proposal** by April 30 that includes:

- ☐ A short description of your planned application
- ☐ A prioritized **list of features** (what must work vs. nice-to-have)
- ☐ A **domain model** represented in **UML**
- ☐ A list of intended technologies and external libraries to be used

 **You must deliver what you propose.** Major changes to your project after approval are not allowed (only small adjustments).

Application Requirements

Your application must include the following:

- ☐ Object-Oriented Programming principles
- ☐ Networking:
 - More than a one-time data request
 - Examples: client-server communication or repeated API queries or Django-based Webserver
- ☐ Database:

- At least 5 interrelated tables
- Must include meaningful data storage, updates, and queries

- ☐ A significant Python backend (web projects must not be frontend-only)
- ☐ Use of technologies from class (e.g., Django, database access, networking)
- ☐ External libraries allowed, such as:

- pygame, pandas, numpy, geographic libraries
- APIs to external media/data sources (e.g., Spotify, YouTube, OpenWeather)

Testing & Documentation

- ☐ You must implement unit/component tests using pytest (covering all critical parts of your application logic) → **This is a significant part of your grade!**

- ☐ Include a **README.md** or **README.pdf** file in your repository that contains:

- A short project description (including context and features)
- Installation & usage instructions (how to run your app)
- How to run the tests
- Overview of features and how to interact with them



Projects that cannot be started will receive a failing grade (= test them on other computers)

Application Submission

Submit your completed project via GitHub Classroom:

👉 <https://classroom.github.com/a/JI7PxpW>

In addition to the source code, make sure to include:

- ☐ requirements.txt (with all dependencies of your project)
- ☐ a database with sample data (so the app is not blank on start)
- ☐ registered users (whichever is applicable)
 - An administrator user (username: admin; password: imc-admin)
 - A “normal” user (username: user1; password: imc-user)
 - (if you have multiple users or classes, add more users)

- ☐ The link to a **YouTube video** (unlisted) of a screen recording showing all functionalities of your app



Submission of all files is a minimum requirement for a passing grade



Ask questions on Teams. Ask early! No assignment questions 72hrs prior to the deadline